Experiment 4 — Deployment of Cloud Infrastructure Components (EC2 + S3 + RDS)

Part A: Compute Resource Deployment (Amazon EC2)

Step 1: Launch EC2 Instance

1. Log in to **AWS Management Console**
2. In the search bar, type **EC2** and click on it
3. On the EC2 Dashboard, click **Launch Instance**
4. Under **Name and Tags**, enter a name (e.g., MyWebServer)
5. Under **Application and OS Images**, select:
   - **Amazon Linux 2** — Free Tier eligible
6. Under **Instance Type**, select **t3.micro** (confirm it shows Free Tier eligible)
7. Under **Key Pair (Login)**:
   - Click **Create new key pair**
   - Enter a name (e.g., my-key)
   - Select **.pem** format
   - Click **Create key pair —** it will download automatically, save it securely
8. Under **Network Settings → Firewall (Security Groups)**:
   - Select **Create security group**
   - Add rules:
     - **SSH** (Port 22) → Source: My IP
     - **HTTP** (Port 80) → Source: Anywhere (0.0.0.0/0)
9. Under **Configure Storage —** leave default (8 GB gp2)
10. Click **Launch Instance**
11. Wait until the instance **State** shows **Running** and **Status Check** shows **2/2 checks passed**

📸 **Screenshots needed:**

- EC2 Launch Instance configuration screen showing AMI, instance type, key pair and security group
- EC2 Instances list showing your instance in **Running** state with its Public IPv4 address

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Part B: Storage Configuration (Amazon S3)

Step 2: Create S3 Bucket and Upload Files

1. In the AWS search bar, type **S3** and click on it
2. Click **Create Bucket**

3. Enter a globally unique bucket name (e.g., myproject-storage-123)
4. Region: **ap-south-1** (same as EC2 — recommended)
5. Keep **Block all public access** enabled
6. Keep **Bucket Versioning** disabled
7. Keep default encryption (SSE-S3)
8. Click **Create Bucket**
9. Click on your newly created bucket to open it
10. Click **Upload** → **Add Files**
11. Select and upload:
    - One **image file** (.jpg or .png)
    - One **JSON file** (.json) — you can create a simple one like `{"name": "test", "status": "active"}`
12. Click **Upload** and wait for success confirmation
13. Click on the uploaded JSON file and note its:
    - Object URL
    - File size
    - Last modified date

📸 **Screenshots needed:**

- S3 bucket list showing your newly created bucket
- Bucket contents showing both uploaded files with Succeeded status
- Object details page of one file showing its URL and metadata

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Part C: Database Deployment (Amazon RDS)

Step 3: Deploy Managed Database

1. In the AWS search bar, type **RDS** and click on it
2. On the RDS Dashboard, click **Create Database**
3. Choose **Standard Create**
4. Under **Engine Type**, select **PostgreSQL** (or MySQL)
5. Under **Templates**, select **Free Tier**
6. Under **Settings**:
    - DB Instance Identifier: **mydb-instance**
    - Master Username: **myadmin**
    - Under Credentials Management: select **Self managed**
    - Set a strong Master Password and confirm it (save it securely)
7. Under **Instance Configuration** — it will auto-select **db.t4g.micro** or **db.t3.micro** (Free Tier)
8. Under **Storage** — leave default 20 GB
9. Under **Connectivity** — leave defaults
10. Under **Additional Configuration**:

    - Enable **Automated Backups** (default 7 days retention)
11. Click **Create Database**
12. Wait until the DB status changes from **Creating** to **Available** (this may take 5–10 minutes)

📸 **Screenshots needed:**

- RDS Create Database screen showing engine selection, Free Tier template, and instance identifier
- RDS Databases list showing your instance with status **Available**

Final Summary:

- Open EC2 and Launch Instance
- Give name, key pair and network settings
- Open S3 and create bucket giving name and region
- Upload pdf or image and view permissions
- Open RDS and create MySQL database
- Give name, password and backups